

LINUX THEORY ASSIGNMENT - 2

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siva@linux:~$ cd Desktop
siva@linux:~/Desktop$ cd permission_lab
siva@linux:~/Desktop/permission_lab$ ls -l
total 4
-rw-rw-r-- 1 siva siva 0 Aug 22 13:42 sample.txt
drwxrwxr-x 2 siva siva 4096 Aug 22 13:42 sample_dir
siva@linux:~/Desktop/permission_lab$ ls -l sample_dir
total 0
-rw-rw-r-- 1 siva siva 0 Aug 22 13:42 test.txt
siva@linux:~/Desktop/permission_lab$ ls -l sample_dir
total 0
-rw-rw-r-- 1 siva siva 0 Aug 22 13:42 test.txt
siva@linux:~/Desktop/permission_lab$ chmod 400 sample.txt
siva@linux:~/Desktop/permission_lab$ ls -l sample.txt
-r----- 1 siva siva 0 Aug 22 13:42 sample.txt
siva@linux:~/Desktop/permission_lab$ cat sample.txt
siva@linux:~/Desktop/permission_lab$ echo "hello">> sample.txt
bash: sample.txt: Permission denied
siva@linux:~/Desktop/permission_lab$ chmod 400 sample_dir
siva@linux:~/Desktop/permission_lab$ ls sample_dir
test.txt
siva@linux:~/Desktop/permission_lab$ cd sample_dir
bash: cd: sample_dir: Permission denied
siva@linux:~/Desktop/permission_lab$ cd ..
siva@linux:~/Desktop$ cd permission_lab
siva@linux:~/Desktop/permission_lab$ chmod 200 sample.txt
siva@linux:~/Desktop/permission_lab$ cat sample.txt
cat: sample.txt: Permission denied
siva@linux:~/Desktop/permission_lab$ echo "hello">> sample.txt
echohello: command not found
siva@linux:~/Desktop/permission_lab$ echo "hello">> sample.txt
siva@linux:~/Desktop/permission_lab$ chmod 200 sample_dir
siva@linux:~/Desktop/permission_lab$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied
siva@linux:~/Desktop/permission_lab$ cd sample_dir
bash: cd: sample_dir: Permission denied
siva@linux:~/Desktop/permission_lab$ touch sample_dir/new.txt

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siva@linux:~/Desktop/permission_lab$ chmod 100 sample.txt
siva@linux:~/Desktop/permission_lab$ cat sample.txt
cat: sample.txt: Permission denied
siva@linux:~/Desktop/permission_lab$ ./sample.txt
/usr/bin/bash: ./sample.txt: Permission denied
siva@linux:~/Desktop/permission_lab$ chmod 100 sample_dir
siva@linux:~/Desktop/permission_lab$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied
siva@linux:~/Desktop/permission_lab$ cd sample_dir
siva@linux:~/Desktop/permission_lab/sample_dir$ chmod 500 sample.txt
chmod: cannot access 'sample.txt': No such file or directory
siva@linux:~/Desktop/permission_lab/sample_dir$ cat sample.txt
cat: sample.txt: No such file or directory
siva@linux:~/Desktop/permission_lab/sample_dir$ echo "hello" >> sample.txt
bash: sample.txt: Permission denied
siva@linux:~/Desktop/permission_lab/sample_dir$ chmod 500 sample_dir
chmod: cannot access 'sample_dir': No such file or directory
siva@linux:~/Desktop/permission_lab/sample_dir$ cd -
/home/siva/Desktop/permission_lab
siva@linux:~/Desktop/permission_lab$ chmod 500 sample_dir
siva@linux:~/Desktop/permission_lab$ chmod 500 sample.txt
siva@linux:~/Desktop/permission_lab$ cat sample.txt
hello
siva@linux:~/Desktop/permission_lab$ ls sample_dir
test.txt
siva@linux:~/Desktop/permission_lab$ cd sample_dir
siva@linux:~/Desktop/permission_lab/sample_dir$ cd -
/home/siva/Desktop/permission_lab
siva@linux:~/Desktop/permission_lab$ chmod 600 sample.txt
siva@linux:~/Desktop/permission_lab$ cat sample.txt
hello
siva@linux:~/Desktop/permission_lab$ ls sample_dir
test.txt
siva@linux:~/Desktop/permission_lab$ cd sample_dir
siva@linux:~/Desktop/permission_lab/sample_dir$ touch sample_dir/new.txt
touch: cannot touch 'sample_dir/new.txt': No such file or directory
siva@linux:~/Desktop/permission_lab/sample_dir$ chmod 700 sample.txt

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NO	Permission	sample.txt — Observation	sample_dir — Observation
1	400 (r--)	It allows Read allowed, modification not allowed	Cannot list contents or enter directory
2	200 (-w-)	It Cannot read; can modify	ThisCannot list, enter, or create
3	100 (--x)	Cannot read; execute permission exists but normal text file won't execute	Cannot list; can enter/traverse
4	500 (r-x)	Can read; cannot modify	Can list and enter; cannot create/delete
5	600 (rw-)	Can read and modify	Cannot list/enter/create because x is absent
6	700 (rwx)	Can read and modify; executable permission available	Can list, enter, create and delete

1. What does read (r) permission allow you to do with a regular file?

When I gave r permission to the file I was able to open and read the contents of the file. I could not modify the file without w permission.

2. What does read (r) permission allow you to do with a directory?

When I gave r permission to the directory I understood that it allows me to list the files inside the directory. I also needed x permission to enter and access the directory properly.

3. What does write (w) permission mean for a regular file?

When I gave w permission to the file I was able to modify or add content to the file. Without r permission I could not read its contents normally.

4. What does write (w) permission mean for a directory?

When I gave w permission to the directory, it allowed me to create, delete, or rename files, but I also needed x permission to perform these operations properly.

5. What does execute (x) permission mean for a regular file?

When I gave x permission to the file it gave permission to execute the file. But since my sample.txt was just a normal text file and not an executable program, it did not run as a program.

6. Why is execute (x) permission important for accessing a directory?

When I tested the directory with only x permission, I was able to enter the directory using cd, but I could not list its contents. So I understood that x permission is required to enter and traverse a directory.

7. Which permission combination allows the owner to fully access a regular file?

700 (rwx) gives the owner read, write, and execute permissions. So the owner can read and modify the file and has execute permission.

8. Which permission combination allows the owner to fully access a directory?

700 (rwx) allows the owner to list and enter the directory and create and delete files inside it.

9. Why do the same permissions behave differently?

While doing the practical, I noticed that the same r, w, and x permissions have different meanings for files and directories. For a regular file, r is used to read the contents, w is used to modify the contents, and x is used to execute the file.